

Securing the Grid: Sequential Anomaly Detection in Industrial Control System Traffic Using LSTM Autoencoders

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Abstract

The increasing integration of digital control systems into power grid infrastructure has introduced significant cybersecurity vulnerabilities, as industrial control system (ICS) protocols such as Modbus, DNP3, and IEC 61850 governing supervisory control and data acquisition (SCADA) networks can be exploited to cause disruptions with severe consequences. Developing robust, automated methods for detecting anomalous network behavior in these environments is essential for strengthening energy security. This study, conducted in the Interdisciplinary Science Department at Brookhaven National Laboratory (BNL), presents the development of a foundational deep learning model for network anomaly detection in power grid communication traffic, focusing on Modbus protocol data extracted from packet capture (PCAP) files. Feature engineering was performed using TShark-based extraction pipelines applied to the CIC Modbus Dataset 2023, which provides labeled examples of both normal operations and various attack scenarios targeting industrial control systems. A Long Short-Term Memory (LSTM) Autoencoder architecture was employed, which leverages the sequential and temporal nature of network traffic to learn normal communication patterns and flag deviations through reconstruction error analysis. The LSTM Autoencoder model achieved a detection accuracy of 99.98%, demonstrating strong capability in identifying anomalous behavior within ICS network communications, and providing a validated unsupervised detection component for network traffic intended to serve as one modality within the broader foundational model architecture. Further work should evaluate the model's generalizability by applying it to additional datasets encompassing different attack types, protocol environments, and network configurations. This study aligns with BNL's role in the Department of Energy's (DOE) Genesis Mission, a national initiative leveraging artificial intelligence to accelerate discovery science, strengthen national security, and drive energy innovation across the DOE's 17 national laboratories. As a result of this research, I have developed a strong understanding of ICS protocol behavior at the packet level, gained hands-on experience with deep learning for cybersecurity applications, and built proficiency in network data processing and analysis using Python, TShark, and machine learning frameworks such as PyTorch.

I. Introduction

The reliable operation of critical national infrastructure, including power generation, water treatment, and transportation networks, necessitates the security of industrial control systems (ICS). Historically, ICS networks were air-gapped and assumed to be physically isolated from external threats. However, the widespread adoption of networked SCADA (Supervisory Control and Data Acquisition) systems and smart grid technologies has fundamentally changed this assumption, exposing vulnerable operational technology (OT) networks to potential malicious actors who previously had no viable entry point.³

High-profile cyber incidents, such as the 2015 and 2016 Ukrainian power grid attacks and the 2021 Florida water treatment intrusion, have demonstrated that ICS cyberattacks can have severe physical consequences.⁴ These events have pushed significant investments into ICS-specific intrusion detection systems (IDS). Although, labeled datasets such as the CIC Modbus Dataset 2023 exists,^{1,2} unsupervised methods offer complementary detection for novel or zero-day attacks not represented in training labels, driving the approach used in this study.

This paper presents an experimental study conducted at Brookhaven National Laboratory (BNL) contributing to the development of a cross-modal anomaly detection foundational model for power grid cybersecurity. The specific contribution is the application and evaluation of an established LSTM Autoencoder anomaly detection approach to the CIC Modbus Dataset 2023, with scope limited to offline analysis of PCAP-format captures. The study evaluates whether reconstruction-error-based anomaly scoring can distinguish normal ICS communication patterns from attack traffic without requiring labeled training data, providing a validated unsupervised detection component for network traffic that is intended to serve as one modality within a broader foundational model architecture.

II. Background and Related Work

A. Industrial Control System Protocols

The Modbus protocol, originally developed in 1979, is a serial communication protocol widely used for polling sensor readings and issuing commands to programmable logic controllers (PLCs). It is dominant in power grid substation environments due to its simplicity and broad hardware support. Modbus polling generates regular, low-latency request-response pairs with characteristic inter-arrival time distributions that serve as a foundation for behavioral modeling.³

B. Machine Learning for ICS Anomaly Detection

Prior work in ICS anomaly detection has applied rule-based systems, statistical process control methods, and, more recently, deep learning approaches.^{5,6} Rule-based systems offer high precision but require manual specification of normal behavior and do not generalize to novel attack vectors. Autoencoders trained exclusively on normal data exploit the intuition that anomalous inputs will produce high reconstruction error.⁷ LSTM-based autoencoders extend this to sequential data and have been applied to ICS anomaly detection in prior work,⁸ making them a well-established candidate for evaluation on Modbus protocol traffic.

III. Methods

A. Dataset

Network traffic was drawn from the CIC Modbus Dataset 2023,^{1,2} which contains PCAP captures from a simulated substation environment including both benign Modbus traffic and nine attack categories: reconnaissance, query flooding, payload loading, delay response, modified length parameters, false data injection, stacking Modbus frames, brute force write, and baseline replay. The dataset was generated using a Docker-based testbed simulating Intelligent Electronic

Devices (IEDs) and SCADA Human-Machine Interfaces (HMIs) communicating over Modbus. Attack scenarios span three configurations: external attacker, compromised IED, and compromised SCADA HMI.

B. Preprocessing

Raw PCAP captures were parsed using TShark and PyShark to extract per-packet fields including timestamps, source and destination IP addresses, transport layer protocol, payload length, and Modbus-specific fields such as function codes and register addresses. Packets were organized into fixed-length sliding windows of 50 consecutive packets per protocol stream to preserve temporal ordering for sequence modeling. Benign traffic windows were used exclusively for training; labeled attack windows were reserved for evaluation.

C. Feature Engineering

The CIC Modbus Dataset 2023 provides 48 extracted network traffic features per flow record. Correlation analysis was performed to identify features with meaningful association to anomalous behavior, yielding 17 candidate features (Figure 1). Further analysis narrowed the final feature set to 5 features most predictive of attack traffic: bit count, reference number, word count, function code, and requests per second (Table 1). These features collectively capture both the structural characteristics of individual Modbus packets and the volumetric behavior of communication sessions, providing a compact but informative representation for sequence modeling. Features were normalized to the range $[0, 1]$ using Min-Max scaling fit on training data only.

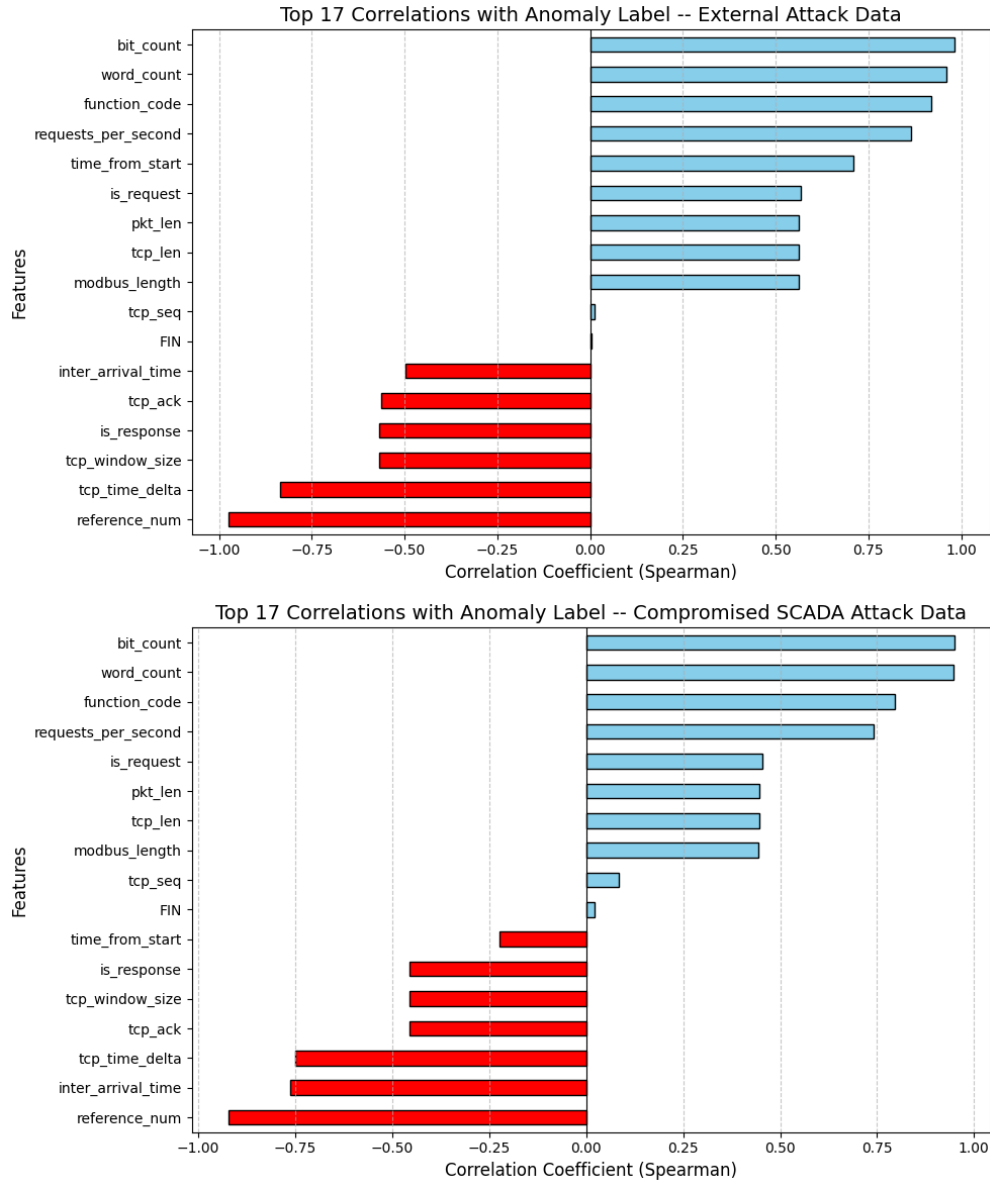


Figure 1: Top 17 Spearman correlation coefficients between extracted features and the anomaly label. (Top) External Attack scenario. (Bottom) Compromised SCADA scenario. Positive values (blue) indicate features more present in anomalous traffic; negative values (red) indicate features more present in benign traffic.

Feature	Spearman Correlation
bit_count	0.9820
reference_num	-0.9755
word_count	0.9607
function_code	0.9201
requests_per_second	0.8642

Table 1: Top 5 features by absolute Spearman correlation with the anomaly label (External Attack scenario).

D. Model Architecture and Training

The anomaly detection model is an LSTM Autoencoder implemented in PyTorch. The encoder consists of a 2-layer LSTM with a hidden dimension of 64, which compresses each input sequence into a fixed-length latent representation. The decoder is a symmetric 2-layer LSTM that reconstructs the original input from the repeated final hidden state of the encoder. Input sequences are sliding windows of 50 consecutive packets, each described by 5 features, yielding an input shape of (batch, 50, 5).

The model was trained exclusively on benign traffic sequences using mean squared error (MSE) as the reconstruction loss. Training used the Adam optimizer with an initial learning rate of 1×10^{-4} , a batch size of 64, and a ReduceLROnPlateau scheduler with patience of 5 and a decay factor of 0.5. Gradient clipping with a maximum norm of 1.0 was applied to stabilize training. The model was trained for 50 epochs with an 80/20 train-validation split on benign data. At inference, the per-sequence reconstruction error was computed, and anomalies were flagged when the error exceeded a threshold set at the 95th or 99th percentile of the training reconstruction error distribution.

IV. Results

The LSTM Autoencoder achieved strong detection performance on both test sets (Tables 2 and 3). The External Attack set yielded a perfect AUROC of 1.0000, while the Compromised SCADA set reached 0.9983. Increasing the threshold from the 95th to the 99th percentile reduced false positives in both cases while maintaining near-perfect recall. Detection proved more challenging on the Compromised SCADA set, where precision at the 95th percentile was 0.80 compared to 0.96 on External Attack, suggesting that compromised SCADA traffic patterns are harder to distinguish from normal behavior. Figure 2 illustrates the reconstruction error over time for both scenarios, with gray-shaded regions indicating true anomaly windows.

Scenario	Precision	Recall	F1-Score	Accuracy	AUC-ROC
External Attack	0.96	1.00	0.98	0.98	1.0000
Compromised SCADA	0.80	1.00	0.89	0.92	0.9983

Table 2: Anomaly detection performance using the 95th percentile threshold (0.0144).

Scenario	Precision	Recall	F1-Score	Accuracy	AUC-ROC
External Attack	0.99	1.00	0.99	0.99	1.0000
Compromised SCADA	0.84	1.00	0.91	0.94	0.9983

Table 3: Anomaly detection performance using the 99th percentile threshold (0.0304).

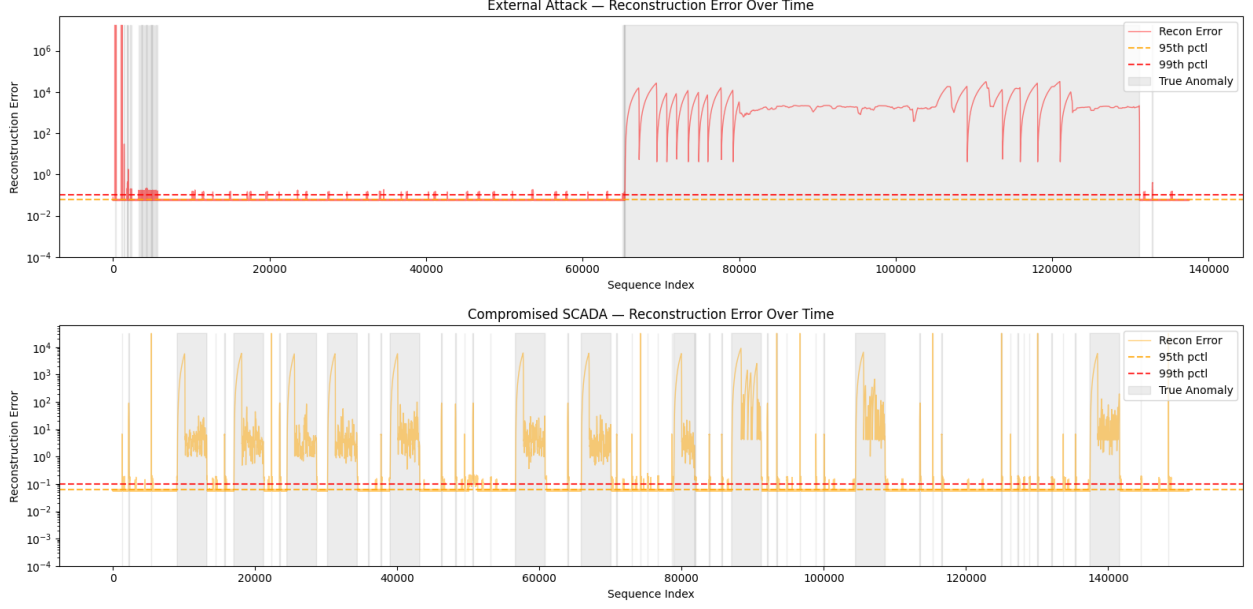


Figure 2: Reconstruction error over time for the (top) External Attack and (bottom) Compromised SCADA test sets. Dashed lines indicate the 95th and 99th percentile thresholds. Gray-shaded regions represent true anomaly periods.

V. Discussion

The results support the hypothesis that LSTM Autoencoders are well-suited for Modbus anomaly detection due to the inherently sequential, time-dependent nature of ICS protocol communications. The model’s ability to learn normal temporal patterns from benign traffic alone and flag deviations at inference time is particularly relevant for operational environments where re-training on new attack signatures is impractical.

Although the CIC Modbus Dataset 2023 includes labeled attack traffic, the unsupervised approach was adopted to evaluate whether anomaly detection is achievable without label dependence, a practically relevant capability for detecting novel attacks outside the training distribution. Several unexplored approaches warrant consideration in future work. Supervised classifiers such as Random Forest or gradient-boosted trees trained directly on the labeled attack categories may achieve higher precision on known attack types and would provide a meaningful performance upper bound for comparison. Graph neural network (GNN) approaches that model device com-

munication topology could capture structural anomalies invisible to packet-level sequence models. Additionally, a hybrid pipeline combining an unsupervised detection layer for novel anomalies with a supervised classifier for known attack categorization may represent the most operationally practical architecture.

Several limitations of this study also warrant acknowledgment. The model was trained and evaluated on traffic from a single simulated environment; generalization to different network topologies or device configurations is unvalidated. The current pipeline operates offline on stored PCAPs, and adaptation to streaming inference would require significant engineering work for latency-sensitive ICS environments.

VI. Conclusions

This study applied and evaluated an LSTM Autoencoder pipeline for unsupervised anomaly detection in Modbus ICS network traffic using the CIC Modbus Dataset 2023. At the 95th percentile threshold, the model achieved an AUC-ROC of 1.0000 and F1 score of 0.98 on External Attack traffic, and an AUC-ROC of 0.9983 and F1 score of 0.92 on Compromised SCADA traffic. At the stricter 99th percentile threshold, performance improved further, reaching an F1 of 0.99 on External Attack and 0.94 on Compromised SCADA, while maintaining AUC-ROC scores of 1.0000 and 0.9983 respectively. The findings confirm that sequence-aware deep learning is a viable approach for ICS intrusion detection, with the model trained without labeled data yet evaluated against ground-truth labels spanning nine attack categories. These results demonstrate complementary value alongside supervised methods, particularly for detecting novel or zero-day attack patterns. Future work should evaluate supervised baselines against the same dataset for direct comparison, and explore hybrid detection architectures suited to real-time substation monitoring.

Acknowledgments

This project was supported in part by the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists (WDTS) under the Community College Internships Program (CCI). I want to thank my mentor, Dr. Meng Yue, for guidance and support throughout this project.

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